

# Autonomous AI Trust Fabric for Clinical Decision Intelligence

*End-to-End Flow Document — v2 (updated with RAG-grounded Explainability)*

## 1. Problem Statement

Hospitals already use diagnostic AI models (vendor-built, black-box) to support clinical decisions — e.g., a model that reviews a chest X-ray and predicts "Pneumonia, 94% confidence." Today, that prediction is just a number and a label. There is no structured, retrievable record of:

- What exact patient data/scan produced this prediction
- Which model version made the call, and whether it was approved
- Why the prediction is clinically reasonable
- Whether the model is behaving normally over time

If a prediction is later questioned (by a doctor, an audit, or a regulator), the answer today is: dig through scattered logs and hope they still exist.

**What we're building:** a system that sits after the diagnostic model and automatically builds a **Digital Compliance Twin** for every prediction — a permanent record combining data lineage, a RAG-grounded plain-English explanation, and a compliance check — so any prediction can be justified at any time.

**We do not build, train, or modify the diagnostic model. It is treated as an external black box,** and its output is simulated with mock prediction records that follow the exact schema a real model would produce.

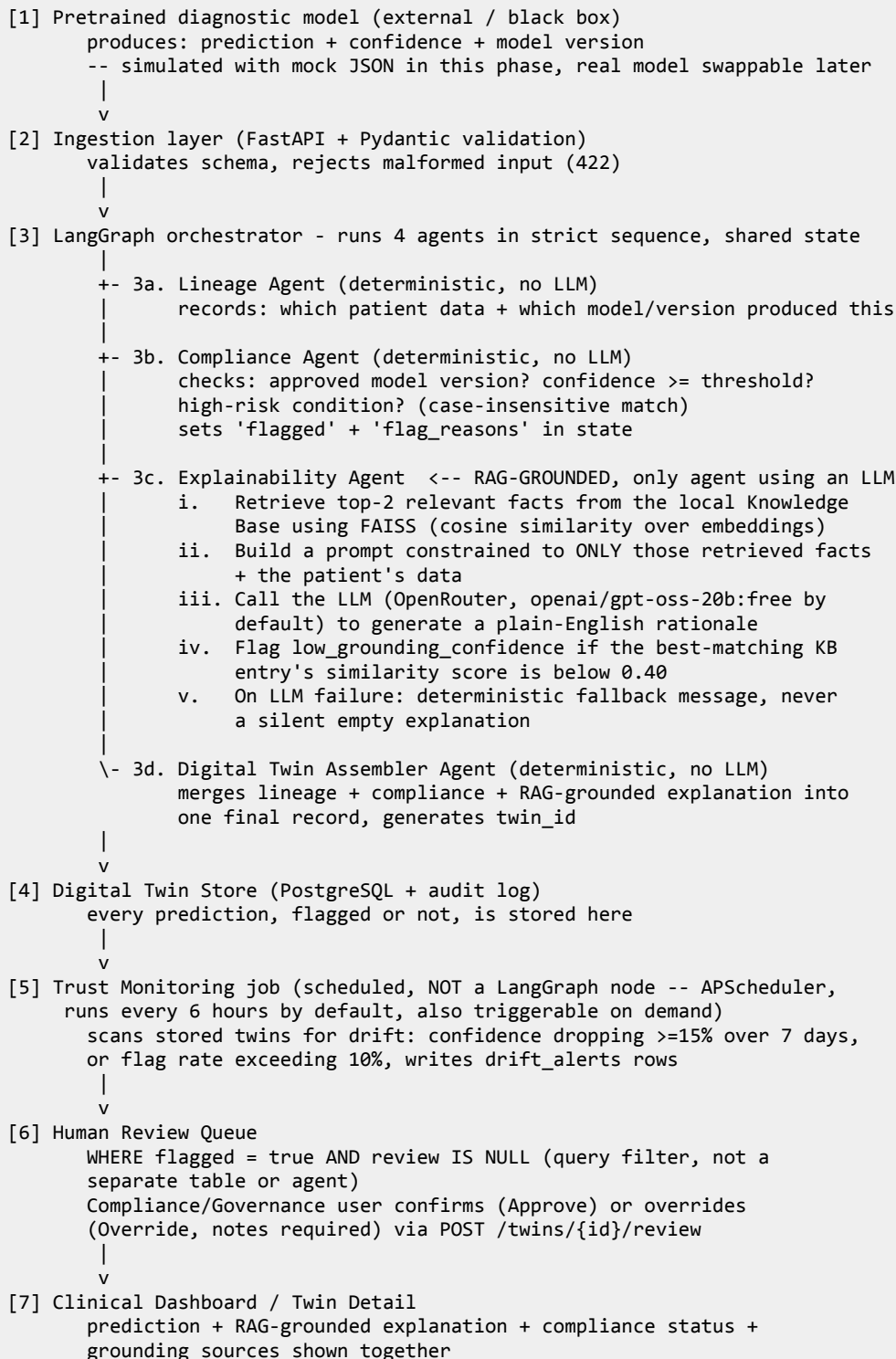
## 2. Users

| Role                                  | Relationship to system                                                                                                                                                           |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Doctors / clinicians                  | Passive beneficiaries — see prediction + explanation in their existing workflow. Read-only access.                                                                               |
| Compliance/Governance (combined role) | Active users — review flagged records, configure the rulebook, and monitor drift alerts. Combines compliance officer, hospital AI governance, and IT admin duties into one role. |

### 3. Business Value

- Reduces liability risk — every prediction has a defensible record
- Speeds up audits — minutes instead of days of log archaeology
- Regulatory readiness — ahead of tightening AI-in-healthcare rules (FDA SaMD, EU AI Act)
- Consistent trust layer across multiple third-party diagnostic tools
- Reduces hallucination risk in AI-generated explanations via RAG grounding — every explanation can be traced back to a specific, citable clinical fact
- Increases clinician trust/adoption via plain-English rationale

## 4. End-to-End Flow



## 5. RAG Knowledge Base — New Component

A small, curated set of clinical reference facts (data/knowledge\_base.json) that the Explainability Agent is constrained to draw from — this is what prevents the LLM from inventing plausible-sounding but unverified clinical claims.

| Layer                | Choice                                                                   |
|----------------------|--------------------------------------------------------------------------|
| Vector index         | FAISS IndexFlatIP (exact cosine similarity search)                       |
| Embedding model      | sentence-transformers/all-MiniLM-L6-v2 (local, no external API call)     |
| Knowledge base size  | 8 curated entries covering Pneumonia, Tumor, Cancer, and Normal findings |
| Retrieval            | Top-2 facts by similarity, computed fresh per prediction                 |
| Low-grounding signal | low_grounding_confidence = true if best match similarity < 0.40          |

Sample knowledge base entry:

```
{
  "id": "kb001",
  "condition": "Pneumonia",
  "text": "Common presenting symptoms of pneumonia include fever, cough (often productive), shortness of breath, chest pain, fatigue, and low oxygen saturation. Risk factors include smoking history, immunocompromised status, recent hospitalization, and advanced age.",
  "source": "CDC Pneumonia Clinical Overview"
}
```

## 6. Input Specification

### 6a. Diagnostic model input (external, simulated)

A chest X-ray produces a mock prediction record, shaped exactly as a real diagnostic model's output would be. No real model is called in this phase.

### 6b. Prediction record (primary input to the system)

```
{
  "patient_id": "P0009",
  "model_name": "vit-xray-pneumonia-classification",
  "model_version": "v2.1",
  "prediction": "PNEUMONIA",
  "confidence": 0.8423,
  "timestamp": "2026-07-31T09:11:32Z"
}
```

### 6c. Patient context (supporting input, per case)

```
{
  "patient_id": "P0009",
  "age": 64,
  "symptoms": ["low oxygen saturation", "cough"],
  "scan_type": "chest_xray",
  "relevant_history": "smoker, prior respiratory issues"
}
```

### 6d. Compliance rulebook (static reference input)

```
{
  "approved_model_versions": ["v2.0", "v2.1"],
  "minimum_confidence_threshold": 0.75,
  "high_risk_conditions_requiring_review": ["Pneumonia", "Tumor", "Cancer"]
}
```

## 7. Sample Output — Explainability Agent (RAG-grounded)

```
{
  "text": "The model's 84.2% confidence pneumonia prediction aligns with
    recognized presenting symptoms - low oxygen saturation and cough are
    both commonly associated with pneumonia, particularly given the
    patient's smoking history and prior respiratory issues, which are
    noted risk factors.",
  "explanation_type": "narrative_llm_rag_grounding",
  "grounded_in_sources": [
    {"id": "kb001", "source": "CDC Pneumonia Clinical Overview", "similarity_score": 0.81}
  ],
  "low_grounding_confidence": false
}
```

## 8. Sample Output — Final Digital Compliance Twin

```
{
  "twin_id": "b3f1...",
  "patient_id": "P0009",
  "prediction": {"label": "PNEUMONIA", "confidence": 0.8423, "model_version": "v2.1"},
  "lineage": {"source": "vit-xray-pneumonia-classification",
    "input_captured_at": "2026-07-31T09:11:32Z",
    "patient_id": "P0009", "model_version": "v2.1"},
  "compliance": {"model_version_approved": true, "confidence_ok": true,
    "high_risk_requires_review": true, "flagged": true,
    "flag_reasons": ["prediction 'PNEUMONIA' is a high-risk
      condition requiring human review regardless of confidence"]},
  "explanation": "...(as above)...",
  "explanation_type": "narrative_llm_rag_grounded",
  "grounded_in_sources": [{"id": "kb001", "source": "CDC...", "similarity_score": 0.81}],
  "low_grounding_confidence": false,
  "flagged": true,
  "twin_created_at": "2026-07-31T09:11:40Z",
  "review": null
}
```

## 9. Tech Stack (Updated)

| Layer               | Tool                                           | Notes                                                              |
|---------------------|------------------------------------------------|--------------------------------------------------------------------|
| Language            | Python 3.11                                    |                                                                    |
| API                 | FastAPI                                        | async, input validation via Pydantic                               |
| Orchestration       | LangGraph                                      | sequential 4-agent graph, no branching except the final flag check |
| LLM wrapper         | LangChain (ChatOpenAI class)                   | pointed at OpenRouter's OpenAI-compatible endpoint                 |
| LLM                 | OpenRouter — openai/gpt-oss-20b:free (default) | swappable to openai/gpt-4o via one env var, zero code change       |
| RAG embedding model | sentence-transformers (all-MiniLM-L6-v2)       | local, no external embedding API call                              |
| Vector index        | FAISS (IndexFlatIP)                            | in-memory, rebuilt from knowledge_base.json at startup             |
| Diagnostic model    | Mock JSON (data/predictions.json)              | shaped like real model output, no real inference                   |
| Storage             | PostgreSQL 16 (Docker) or SQLite fallback      | twin records + audit log, dialect-agnostic ORM                     |
| Scheduling          | APScheduler (in-process)                       | Trust Monitoring job — Airflow not justified at this scale         |
| Frontend            | React 18 + TypeScript + Vite + Tailwind CSS    | dashboard + review queue + rulebook + trust monitoring             |
| Auth                | Username + password -> role lookup             | no JWT, no Keycloak in this phase                                  |
| Containerization    | Docker + docker-compose                        | Postgres only                                                      |

## 10. Where DataOps Fits

Not a separate layer — embedded directly in the pipeline:

- Input validation (Pydantic schema) = data quality gate
- Lineage Agent = data lineage tracking
- Trust Monitoring job = pipeline/model observability (confidence drift + flag-rate drift)
- RAG grounding = a data-quality control specifically for the explanation layer — constrains LLM output to verifiable source facts, catching potential hallucination risk before it reaches a clinician

## 11. Explicit Scope for This Phase (Simulated vs Real)

| Component                   | Status this phase                                         |
|-----------------------------|-----------------------------------------------------------|
| Diagnostic model prediction | Simulated (mock JSON, shaped like a real model's output)  |
| Patient context data        | Simulated seed data (60 patients)                         |
| Compliance rulebook         | Real, hand-authored, editable via UI                      |
| Explainability              | Real — RAG retrieval (FAISS) + real LLM call (OpenRouter) |
| RAG knowledge base          | Real, but small and hand-curated (8 entries)              |
| Message queue               | Not implemented — direct synchronous call instead         |
| Scheduler                   | In-process (APScheduler), not Airflow                     |
| Cloud hosting               | Local only                                                |
| Auth                        | Hardcoded username/password, no JWT/Keycloak              |
| FHIR / real EHR integration | Not implemented                                           |